



19 SEPTEMBER 2026

Data Centers

Land, iron, heat — orbit still a filing

Snapshot · Markets · Canada · SpaceX / Terafab · Flight proof
NVIDIA · Amazon · Microsoft · Heat · Pushback · Alternates

Roy Poystila & Grok · Announced gigawatts are not live megawatts.

A note on this briefing

This briefing is a collaboration between **Roy Poystila** and **Grok (xAI)**. Roy set the questions and kept folding in what the last draft had left thin — neighbors, orbit and mines, SpaceX, heat, Canada, the companies that draw the rack. Grok did the public-source research, the synthesis, the charts, and the file. Neither side wrote it alone. Where the text is a reading rather than a measurement — what a filing is worth, which escape from a queue will matter — that judgment is Grok's, shaped by the questions Roy kept asking.

It is a working research briefing as of 19 September 2026. It is not an offering document, not an audit of any company's books or construction site, and not a forecast to trade on. Announced gigawatts are not live megawatts. First-power and first-launch dates slip. Capital-expenditure (capex) figures are guidance ranges. Operator claims on water, households heated, or tokens per megawatt are repeated as claims.

If a sentence is a fact in the public record — a Federal Communications Commission (FCC) file number, a named power-purchase agreement (PPA), a published rack kilowatt, a provincial cap — treat it as reportage subject to the next U.S. current report (8-K). If a sentence is about what a player is optimizing for, treat it as structured opinion inside the collaboration: a map, not a verdict.

What a later critique added — without deleting the rest

A shorter restatement of this file was right about three unglamorous facts: the big electrical transformers take three to four years to get; “warm water cooling” loses punch on a hot afternoon unless you use water or slow the computers down; and a satellite computer has to die safely on the way back to Earth. Those points are in the body, in industry language, with the filings attached. The long argument is still here. Sections run 1–14 in order.

Prepared jointly by Roy Poystila and Grok. Revise when the facts move.

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Bottom line

Two tracks, and what should survive the next earnings season

Glossary

Shop language, written out

Sources

Filings, outlooks, disclosures

How to read the file. Tables and file numbers are for specialists. The sentence before the table is for everyone. An acronym is written out the first time; the glossary repeats the expansions. If a line says “talked as” or “if the plan holds,” it is a reading, not a measurement.

The picture in brief

What a data center is for

Before the queues and the racks, here is the job. A data center is a building that keeps other people's computers running without pause: power, cooling, network, physical security, and someone awake at 3 a.m. if a pump fails. Almost nothing you already treat as "the internet," "the cloud," or artificial intelligence (AI) lives on a laptop. It lives in one of these rooms.

What they provide, in industry terms, is leased or owned compute and storage with a contract attached. Banks clear payments there. Hospitals keep records there. Retailers run inventory and checkout there. Governments keep registries there. Streaming, search, maps, and messaging are the consumer face of the same rooms. Colocation landlords rent the room and the megawatts. Cloud companies rent you a slice of the computers. Enterprises still run the rooms they refuse to give up. Edge sites stash a copy of content close to a city so a video does not have to cross an ocean.

The AI wave did not invent the building. It changed what the building is asked to do. Training a frontier model is a weeks-long industrial process that wants cheap firm power and does not care if you are twenty milliseconds away. Inference — answering you, now — wants to sit near users and data. That is why the same industry now sells two products that look like one: a factory in a field, and a hall on a metro fiber ring. Everything that follows is about those products colliding with power, heat, money, and neighbors.

The internet, the cloud, and the current wave of AI all live in buildings that eat electricity and throw off heat. Those buildings are multiplying. The simple version of the story is that the world needs many more of them, fast. The true version is that the world can announce as many as it wants. It can only switch on the ones that already have power equipment, a cooling design that works in that climate, and a town or a province willing to live next door.

Start with scale. There are on the order of eleven to twelve thousand data centers on Earth. Most are ordinary. The ones that matter for AI are the factories: a few hundred large campuses, and a pipeline of several hundred more on paper. Global electricity demand from this sector is estimated near 161 gigawatts in 2026, up from about 123 last year. Some compounding forecasts wave at five hundred gigawatts by 2030. Grids, turbine factories, and transformer plants cannot deliver that. Treat the high number as a warning, not a construction schedule.

The industry is splitting in two. One track is already pouring concrete next to gas plants and existing substations — Meta in Louisiana and Alberta, Amazon in Indiana, Microsoft pairing new halls with a nuclear restart and a West Texas gas plant, xAI filling old industrial shells in Memphis. The other track is a press release waiting on a permit, a queue, or a piece of electrical equipment that takes three to four years to manufacture. A campus beside a power plant still sits dark if the transformer was not ordered years ago.

What goes inside the building is changing as fast as the map. NVIDIA still draws the rack that most halls are built around. Those racks have gone from tens of kilowatts to one or two hundred, with drawings for six hundred and talk of a thousand. Cooling has moved from blowing air to running warm water across the chips. That works cleanly in a Nordic winter, where the leftover heat can warm apartments. It is a different problem in Texas or the Gulf, where the outside air is already hot and the water fight comes back. Halls built for last year's rack can become short-lived industrial boxes if they cannot be replumbed.

Communities have noticed. Virginia taught the industry that a power line has voters. Texas is auditing a huge interconnection queue after winning the last sprint. Scotland paused new hyperscale consents in parliament. Alberta will take the megawatts only if the operator brings its own plant. This is not a sideshow. It is why the next campuses show up next to generation, in secondary metros, or — on paper — under the sea, in mines, on barges, and in orbit.

Orbit is the sentence that gets more attention than it has earned as supply. SpaceX has a real cluster on the ground, a million-satellite filing called Starlink, and a Texas fab idea called Terafab. Starship is still proving it can be a truck. A graphics processing unit (GPU) has flown in orbit at kitchen-appliance power, thermally limited. Radiation shielding fights the safety rule for what may survive re-entry. Read that chapter as engineering and regulation, not as next year's training fleet.

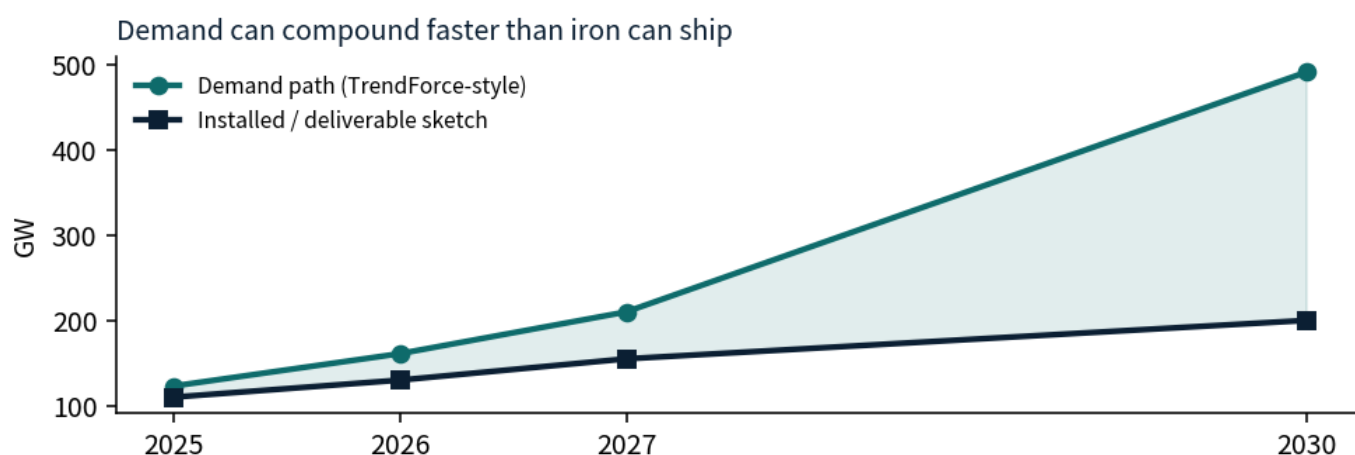
If you want the short map: power and electrical iron decide where a hall can live; the chip rack decides what the hall must be; politics decides whether it opens; fiber, export licenses, and insurance decide which of the permitted sites are actually usable. Everything else — nuclear contracts, custom chips, underwater pods, satellites — is an attempt to escape someone else's queue. Gas and today's racks are the escapes already running. Remote power is mostly a training map; live answers still want the old fiber cities. Some of the gigawatts in official queues are phantom — the same cluster filed more than once. The rest will decide whether 2030 is a thicker 2026 or a different industry.

1. Global snapshot

Start with the obvious question: how big is this, really? Walk through a directory and you will count eleven or twelve thousand data centers. Walk the power bills and you will find that most of those sites barely move the total. Gigawatts, not buildings, are the honest unit — and even those gigawatts split into what is live and what is only asked for.

Data centers are the physical plant of cloud, the public internet, and the AI build. As of 2026, directories put the world in the 11,000–12,000 facility range. Count is a weak proxy: a retail colo cage and a multi-gigawatt AI campus both count as one site. Information-technology (IT) load — the electricity used by the computers themselves — in gigawatts is the number that matters.

Global demand sat near 123 GW in 2025 and is estimated near 161 GW in 2026, with AI servers moving from about a quarter to about a third of draw. Directory counts of 11,000–12,000 sites mix a retail cage with a multi-GW factory. Synergy-style hyperscale tallies are the better object: on the order of 1,300 operating hyperscale sites and ~770 in the announced pipeline. That pipeline is a ceiling, not a construction schedule. Analysts talk about ~100 GW of new installed capacity this decade. Compounding demand cases in the TrendForce family push toward ~491 GW by 2030 if AI load does not break. Those two numbers cannot both be operational facts. The gap is turbines, transformers, hearings, and water.



Demand path versus a sketch of what grids and factories can energize. The 2030 demand figure is a compounding case, not a budget.



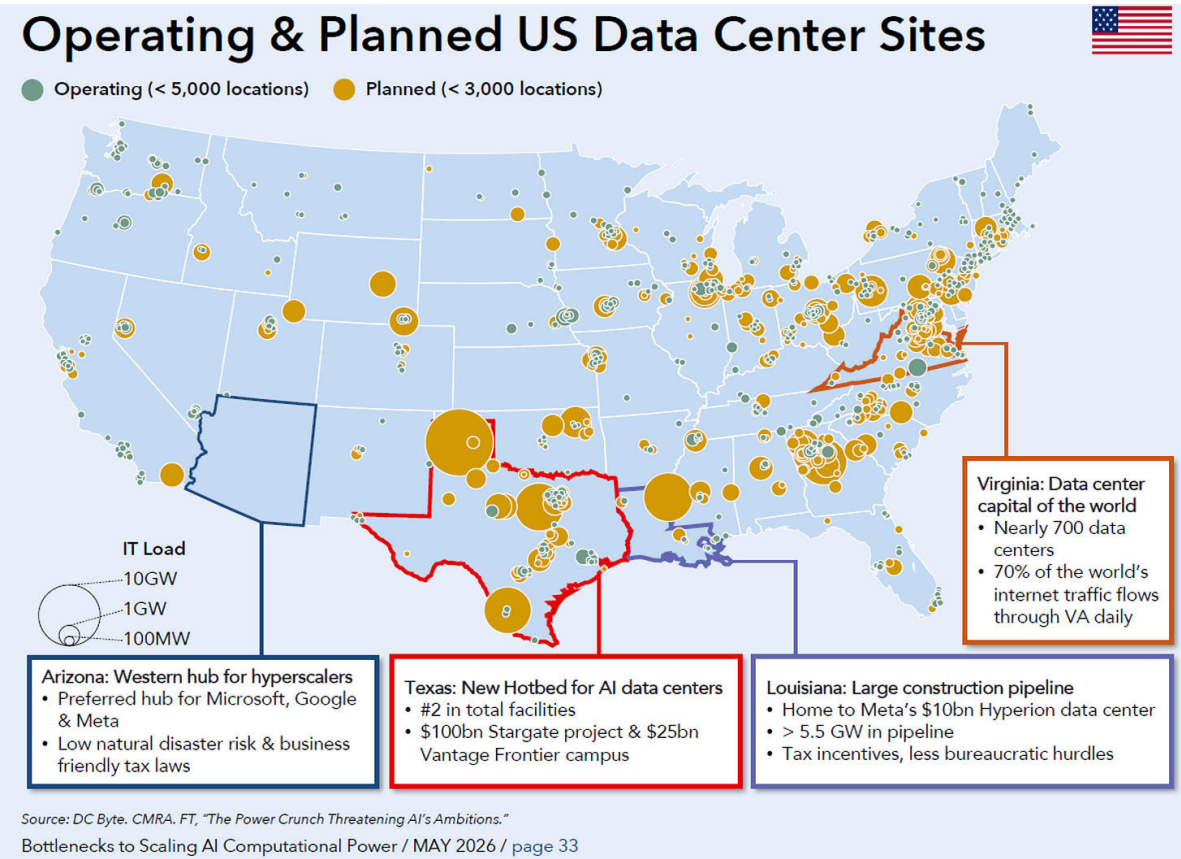
Hyperscale campus at warehouse scale — the object counted as “one site.”



Another factory-class hall. Facility count is a weak proxy for gigawatts.

Country / region	Approx. sites	Strategic note
United States	4,000–4,400	Dominant in count and gigawatts. Northern Virginia still the interconnection capital; Texas the AI-campus sprint.
United Kingdom	~500–555	London/M4 core tight on power; pipeline shifting north.
Germany	~500–523	Frankfurt is the continental hub; efficiency mandates now bite.
China	~370–450 dir.	Directory count understates large state and regional cloud hubs.
Canada	~285–290	Live buildings in Ontario, Quebec, and British Columbia; megawatt pipeline has moved to Alberta gas.
France, India, Australia, Japan	mid-hundreds each	Sovereign and cloud builds; grids and land politics differ by metro.

The United States is not a rounding error on the map. It is also where the political and electrical constraints showed up first. Europe’s problem is similar with less gas slack. Asia’s problem is a mix of power, land, and — for training-class clusters — export rules on the chips that fill the hall.



U.S. concentration is the point of the map. It is also why U.S. hearings now set the tone for siting everywhere else.

2. Usage categories

Not every building in the directory is doing the same job. A bank’s basement, a landlord’s colo hall, and a multi-gigawatt AI factory share a name and almost nothing else. The buckets below are how the industry tells them apart. The twist is that even “AI” splits: the campus that trains a model can sit next to a gas plant in the prairie. The campus that answers you in real time still wants to live in a fiber city.

The industry still uses five working buckets. They now describe different physics, not just different leases.

Type	Who	What it now requires
Hyperscale / AI factory	Amazon Web Services (AWS), Azure, Google, Meta, Oracle, xAI	100 megawatts to multi-gigawatt campuses; high-density liquid cooling; increasingly own generation rather than a utility queue.
Wholesale / retail colo	Equinix, Digital Realty, Vantage, QTS	Landlord sells a powered shell. Wholesale blocks leased to hyperscalers dominate new GW. Time-to-power is the product.
Enterprise	Banks, governments, corporates	Still the majority of site count. Share of new megawatts is falling as workloads move to cloud and AI factories.
Edge / micro	Telco, content-delivery network (CDN), regional cache	Latency sites at the metro edge. Not where frontier training lives.
Sovereign / high-performance computing (HPC) factory	National labs, Gulf funds, EuroHPC	Same liquid halls and rack drawings, different owner, export rules, and political mandate.

The buckets still earn their keep if you refuse to treat them as equal. **Enterprise** still wins site count — banks, governments, and corporates run thousands of rooms — and is losing the new-gigawatt race. Those rooms are not being demolished. They are being capped while training and large inference move to someone else’s campus. Counting sites and counting megawatts now tell opposite stories.

Wholesale colo is no longer a real-estate product with power attached. Time-to-power is the product. Equinix, Digital Realty, Vantage, QTS and their peers sell a shell that already has a feeder, a transformer slot, and a cooling plant that can take a 120–200 kW rack. Hyperscalers still lease those blocks when their own plant is late. That is swing capacity, not a competing theory of the industry.

Hyperscale / AI factory is the bucket that changed the physics. A 100 MW–multi-GW campus is specified backward from a liquid rack, often sits next to generation, and is staffed like a power plant that happens to make tokens. **Edge** still exists for latency — CDN, telco, metro cache — and is irrelevant to frontier training. **Sovereign and HPC factories** use the same drawings under different owners: EuroHPC, Gulf funds, national labs. Export rules and political mandate change. The cold plate does not.

The migration that matters for 2026–30 is not enterprise-to-cloud. It is cloud-to-factory: a handful of owner-operators pouring multi-GW campuses next to generation, while colo remains the swing capacity for whoever cannot wait on their own plant.

Training can leave town. Inference mostly cannot.

Treat “AI load” as two jobs or the map lies. **Training** — the multi-week run that builds or heavily updates a frontier model — does not care whether the user is 20 milliseconds away. It cares about cheap firm power, liquid cooling, and a fabric inside the hall. That is why Hyperion, Rainier, West Texas gas, and Alberta can work as training geography. **Inference** — answering a live query — is bound by data gravity: the model wants to sit near the data and the user, on dense fiber, in Ashburn, Dallas, Dublin, Frankfurt, Tokyo. A 200 ms round trip to a prairie campus is fine at the end of a two-week training run. It is a product failure for a copilot.

Off-grid gas, remote acreage, and nuclear co-location are therefore a *training-first* escape, not a universal one. Some batch inference and fine-tunes can ride along. Real-time enterprise and consumer serving still pulls gigawatts back into the old corridors — which is why NoVA and FLAP-D do not empty out just because Texas and Alberta are pouring factories. The two maps stack. They do not replace each other.

The cable, the license, and the policy

Cheap power is not the same as being on the network. Inference needs dense fiber, a peering hotel or internet exchange, and a path that does not add forty milliseconds. A prairie campus can be electrically perfect and still be the wrong building for a copilot. “Near a city” is not the same as “on the right cable.” Carrier hotels in Ashburn, Dallas, Dublin, Frankfurt, and Tokyo are why those names keep recurring after the gas map has moved on.

A second filter sits above the dirt: who is allowed to run which chips, on whose soil, under whose audit. United States export rules on advanced accelerators, end-use checks, and incoming investment review decide whether a frontier cluster may legally exist in a given country. Data-residency and privacy law pull enterprise inference and government workloads toward national clouds even when Texas power is cheaper. That is why Gulf and European “sovereign” campuses are not only prestige projects. They are license-and-jurisdiction buildings wearing a cooling plant.

3. Primary markets — and the iron that is not a turbine

If you only remember one market, you will remember the wrong lesson. Northern Virginia is still the biggest interconnection hub on Earth — and the place that taught the industry a feeder has voters. Texas is the sprint. Europe is the squeeze. The Gulf is buying into heat. And underneath every map pin is a transformer that may not exist yet. The rows below are bottlenecks with place names, not a tour.

Northern Virginia (“Data Center Alley”) remains the largest interconnection market on Earth and the cautionary tale. Multi-gigawatt operational inventory sits on a grid that is now a political object: retail rates, data-center riders, and transmission upgrades that households can see on a bill. Prince William’s Digital Gateway fight is the emblem, not an outlier.



Northern Virginia inventory at ground level: the largest interconnection market, and the place that taught the industry that feeders have voters.

Texas is the fastest AI-campus market because land is cheaper and Electric Reliability Council of Texas (ERCOT) interconnects have been faster than PJM Interconnection (the Mid-Atlantic / Midwest U.S. grid) — which is exactly why the state is now auditing a tens-of-gigawatt queue and why towns such as San Marcos have reached for local bans. West Texas adds behind-the-meter gas. Dallas–Fort Worth and Austin add metro fiber. The combination is why so many named 2026 campuses have a Texas address.

The European core often called FLAP-D — Frankfurt, London, Amsterdam, Paris, Dublin — is still under grid and water stress at once. New load is leaking north and east. The Nordics sell something Virginia cannot: warm-water heat into a city network. The Gulf is buying sovereign campuses into the worst ambient-heat climate on the map, which is a cooling problem before it is a capital problem.

Market	Role	Binding bottleneck
Northern Virginia	Global interconnection capital	Transmission; household rate backlash; land-use fights
Texas (ERCOT)	AI mega-campus expansion	Queue audits; municipal bans; summer thermal derating
FLAP-D	European core	Grid moratoria; power and water permits
Nordics / Helsinki	Heat-reuse hubs	Fiber egress; smaller local markets
Gulf (Abu Dhabi / UAE)	Sovereign AI bets	Ambient heat; water; geopolitics
Singapore / Tokyo	APAC interconnection	Power caps, land, and (Singapore) a long pause then tight quota



The unglamorous constraint: generator-step-up (GSU) transformers and switchgear next to the hall. Lead times of three to four years. A campus without this iron is a warehouse.

Substation iron is now a co-equal bottleneck with turbines

Gas-turbine order books sold out into the early 2030s are widely tracked. Less tracked: high-voltage GSU transformers and medium-voltage switchgear with manufacturing lead times of three to four years. Siting a hall next to a wellhead or a restarted reactor solves fuel. Without a transformer that was ordered in 2023 or 2024, the hall is a warehouse. This is why “behind-the-meter” is not a slogan. It is a procurement calendar that starts before the groundbreaking photo.

What the queue does not show

The interconnection list is not the whole risk file. Flood plains, wildfire, hail, and a rising count of extreme-heat days are now insurance and lender questions. Some counties still have power and land and are becoming expensive or uninsurable. A campus that cannot buy property cover on terms a bank will accept does not close, even if the feeder is real.

The rest of the bill of materials slips on its own clock: chillers and coolant-distribution units, busbars, generators, batteries, fire-suppression systems a marshal will sign for a liquid hall, copper. Those queues are quieter than GPUs and turbines. They still move first power. Labor is the last unglamorous line. A liquid-cooled, high-voltage plant in a prairie county still needs people who have run one. That talent still clusters in the old metros.

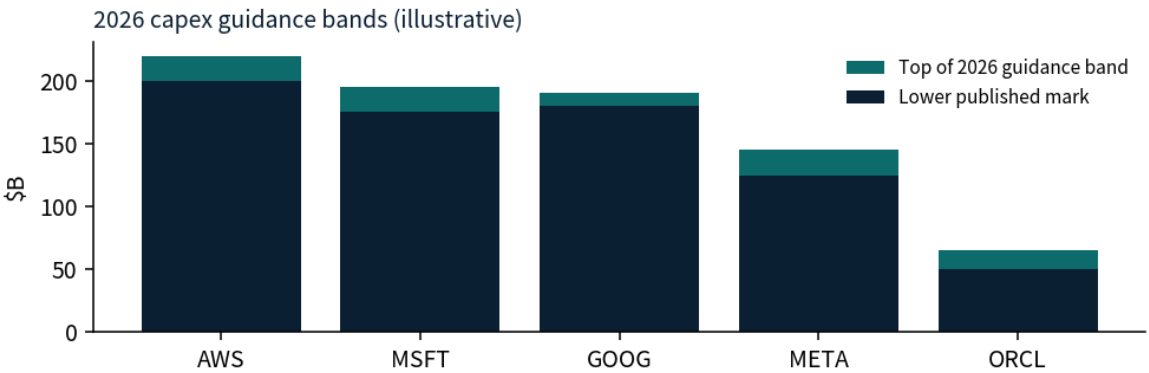
Two demand-side twins belong in the same picture. Software efficiency — distillation, smaller models, more tokens per watt — can make some announced gigawatts optional if serving gets cheaper faster than halls open. And the interest-rate cycle changes the same campus: fifteen-year paper against a three-year rack is a different asset at 3 percent than at 7. Carbon accounting and air permits sit beside all of this. A gas plant next to the hall needs nitrogen-oxide headroom. A European campus may owe a heat-reuse assessment. Cheap renewable certificates are not the same as power on the feeder at 4 p.m. in July.

4. Pipeline and hyperscaler capex

Follow the checks, then follow the dirt. The large operators have told markets they will spend on the order of three-quarters of a trillion dollars this year. Named campuses — Hyperion, Rainier, Fairwater, Stargate — are where that money is supposed to become a hall. Between the guidance and the groundbreaking sits a quieter inflation: queue positions filed more than once, for load that may never show up.

2026 capex guidance across the top operators sits in a \$700–850 billion band depending on who is counted as a “top” name. Spend is outrunning operating cash flow at several firms; debt and project finance are part of how halls get poured. Treat every figure below as guidance, not audited concrete.

Operator	2026 guidance	What the money is pointed at
Amazon / AWS	~\$200B+	Project Rainier (Indiana); Trainium 2/3; owned campuses plus leased regions worldwide.
Microsoft	~\$175–195B	Fairwater-class campuses; OpenAI infrastructure; Crane / Three Mile Island nuclear PPA; West Texas off-grid gas with Chevron.
Alphabet (Google)	~\$180–190B	Tensor Processing Unit (TPU) v6/v7; Visakhapatnam with Adani; on-site turbines and fuel cells after a late start on gas.
Meta	~\$125–145B	Hyperion (Louisiana, multi-GW); Prometheus; El Paso; Sturgeon County; Meta Training and Inference Accelerator (MTIA) in the same rack.
Oracle	sharp step-up (~\$65B class)	Stargate / Abilene with OpenAI and Crusoe; 50,000-GPU AMD Helios cluster as dual-source proof.
xAI / SpaceX compute	cluster spend + filings	Colossus / Colossus II Memphis; Starmind FCC file; Terafab framework in Grimes County.



Illustrative 2026 guidance bands from public ranges. Not audited spend.

Named projects worth tracking as physical objects

Meta Hyperion (Richland Parish, Louisiana) is the template. Public descriptions put it in the multi-GW class — Hyperion has been talked near 5 GW over phases — sitting next to on-site gas so the campus does not wait out a transmission queue. The nuclear PPA book (Vistra, Oklo, TerraPower, Constellation and peers) is the 2030s story, not the 2026 energization story. Prometheus, El Paso, and Sturgeon are the same idea in other dirt: factory next to generation, not a region on a cloud map.

AWS Rainier (New Carlisle, Indiana) is the Anthropic-weighted training campus and a test of whether a Midwestern grid node can absorb factory-scale load without becoming Northern Virginia. It is also the tell that Amazon will own the hall when the workload is frontier training, and still lease colo when the workload is regional inference. Amazon’s Trainium custom chips in the same complex are how AWS tries not to pay list price on every token those halls serve.

Microsoft Fairwater plus **Crane / Three Mile Island Unit 1** are a pair, not two stories. Fairwater is the liquid hall. Crane is the first named nuclear electrons in the hyperscaler book, talked for around 2027. The Chevron West Texas behind-the-meter gas plant (~2.7 GW class in public descriptions) is how Microsoft covers the years before those nuclear electrons exist. Read them together or you will think Microsoft solved power. It sequenced it.

Stargate is two bets that share a name. Texas — Oracle, OpenAI, SoftBank, Crusoe, Abilene-class dirt — is speed on U.S. gas and existing interconnect muscle; first clusters have been talked for late 2026. Abu Dhabi with G42 is sovereign capital buying a training map that does not have to ask PJM. Full announced gigawatts on both sites sit well beyond first power. **Google–Adani Visakhapatnam** is the same move in India: a named AI hub attached to a local energy partner, not a leased hall in Mumbai.

A large slice of every named pipeline is still a queue position, a transformer slot, or a hearing. Read announced GW against Data Center Watch-style delay tallies. Early-2026 U.S. tracking already had on the order of 75 projects blocked or delayed in a single quarter. Pipeline math that ignores that filter is a fundraising slide.

Phantom load — the queue is not a census

Utilities have a name for the rest of the inflation: phantom load. Capgemini’s June 2026 survey of more than 600 electricity executives (*AI meets the grid*) found that 67% see speculative data-center requests, and that about 19% of those requests never become live megawatts. That is not “one in five of every watt on Earth.” It is one in five of the requests the surveyed utilities already treat as suspect. Developers also shop the same underlying server allocation across several utilities and provinces at once; Federal Energy Regulatory Commission (FERC) commissioners in 2026 described that shopping as clogging studies and double-counting expected load. When one of those twins withdraws, the remaining queue often does not simply move up. The study was done for a specific load at a specific bus. A restudy can take six months to more than a year and can change the upgrade bill.

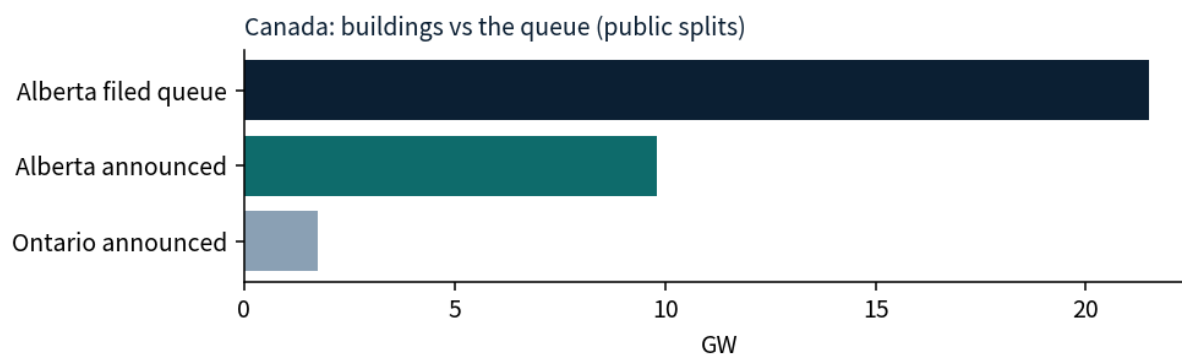
That is why a 21.5 GW Alberta interest queue or a 50 GW-class ERCOT book cannot be read as future construction. Some of it is real. Some of it is optionality. Some of it is the same cluster filed three times. Grid planners who cannot tell the difference freeze or restudy, and the live project waits behind the ghost. Behind-the-meter gas is, among other things, a way off that particular treadmill.

5. Canada — buildings in the east, megawatts in Alberta

Watch one country and the whole argument shows up in miniature. Canada already has the buildings in Ontario, Quebec, and B.C. — hydro, fiber, the old map. The new megawatts have filed in Alberta, next to gas. The bargain that lets any of it proceed is blunt: bring your own plant, accept the cap, pay the levy. If you understand that split, you understand why “clean power” is not the same as “spare power.”

Canada is the cleanest split-screen in the briefing. Live facility stock sits in Ontario (about 110 sites), Quebec, and British Columbia — the old hydro-and-fiber provinces. Something like nine-tenths of the proposed megawatt pipeline has filed in Alberta. One public split of that book: on the order of 21.5 GW in the filed interest queue against about 9.8 GW announced, versus about 1.75 GW announced in Ontario.

The Alberta Electric System Operator (AESO) 1,200 MW interim large-load allocation is already spoken for: Meta’s Sturgeon County campus at 970 MW and Keephills Phase I at 230 MW. Meta’s roughly \$13 billion Sturgeon project is the first Canadian hyperscale of this cycle, paired with the adjacent ~\$4.6 billion, 932 MW Greenlight gas plant. That pairing is the whole Canadian sentence: you may have a grid allocation, and you still bring your own plant.



Public splits of announced versus filed Canadian load. Live buildings remain in Ontario, Quebec, and B.C.

Alberta's working political bargain is bring-your-own-power, a 120-day shot clock for qualified projects, and an equipment levy designed so household rates do not underwrite compute. The quiet constraint behind the gas story is pipes, not only turbines: feeding on the order of 6 GW of planned on-site generation implies billions in new gas-pipeline capacity. Ontario still has the buildings and three-to-five-year connection waits. Quebec's earlier pause and Bill 69 — legislative sign-off for loads above 5 MW — plus B.C. large-load caps mean hydro provinces are structurally harder to enter than gas Alberta, because clean power is already promised to decarbonizing industry. Canada is not short of electrons in the abstract. It is short of electrons that are allowed to become a data-center feeder this decade.

6. SpaceX — Colossus on the ground, Starmind in orbit, Terafab for the silicon

Headlines treat SpaceX compute as one story. It is three, stacked, and only one of them is renting time. On the ground: Colossus in Memphis. On paper in orbit: Starmind. In Texas filings: Terafab, a fab that would try to make the chips instead of standing in Taiwan Semiconductor Manufacturing Company's (TSMC) line. Keep them separate or you will award orbit a gigawatt it has not flown.

SpaceX is running three stacked compute stories, not one. They get collapsed in headlines. They should not be.

Ground: Colossus

xAI's Memphis clusters are the terrestrial speed run: industrial shells converted in months, tens of thousands of GPUs, gas on the fence, early reliability constrained by backup power and the usual sins of moving fast. That is the only layer already renting compute at scale. Google as a large renter of that capacity is the tell that even rivals will buy time on a messy cluster if the alternative is a queue. Colossus is not a science project. It is a hyperscaler with worse uptime dogma and cheaper incremental megawatts.



Starship stacked at the pad — the vehicle Starmind is written as if it were already a bus. It is not.

Orbit: Starmind

In January 2026 SpaceX filed FCC application SAT-LOA-20260108-00016 for an Orbital Data Center system of up to one million satellites. The first-generation AI1 craft is specified near 20 m tall, 70 m deployed wingspan, up to about four tonnes, about 120 kW average and 150 kW peak compute, ~150 kW solar, and ~110 m² of deployable liquid radiators. Results are meant to come home over Starlink lasers. First prototypes are talked for late 2027, with volume after a Gigasat-class production line. Company remarks have put 100 GW in orbit at “on the order of 10,000 Starship launches.” The S-1 is colder than the keynotes: the orbital plan rests on unproven technologies, or technologies that do not exist.

Silicon: Terafab

Terafab is the attempt to stop buying the next increment from the same TSMC / NVIDIA queue every other hyperscaler is in. Texas filings and 2026 announcements describe a Grimes County campus talked as on the order of 100 million square feet across phases, covering mask design, logic and memory, and advanced packaging. Intel is the process partner — 18A in some early descriptions, 14A as the full-scale node, with the 14A process-design kit (PDK) that outside designers need not expected until late 2026. Construction start has been listed around 1 December 2026, with first-phase completion targets spoken in the 2028 window. Headline dollars run from the mid-tens of billions to a \$119 billion four-phase envelope. SpaceX’s own S-1 registration statement still calls Terafab a general framework without binding commitments that Tesla or Intel remain indefinitely. A Western District of Texas trademark action over Terafab / TERA-print is the legal noise around the name, not proof the fab exists.

Musk’s public binary — build Terafab or fail to scale — is strategy. Outside analysts have described the same project as closer to an Intel foundry expansion with Tesla / SpaceX / xAI as anchor customers than a greenfield integrated device manufacturer. Both readings can be true at once. Without radiation-tolerant logic, memory, and packaging in-house, an orbital compute fleet is just another purchase order against TSMC’s Chip-on-Wafer-on-Substrate (CoWoS) packaging line.

6b. Flight requirements — proven versus planned

Now the unromantic question: what has to be true for that orbital filing to become a machine? Starship has to be a truck, not a test article. Heat has to leave a satellite that cannot open a window. Shielding has to protect the chips without surviving re-entry as a weapon. A million-sat application is a claim on all of that. It is not a fleet.

Starwind is written as if Starship were already a bus route. Through mid-September 2026 it was not. Only two Starship flights had flown in 2026 before Flight 14. Flight 13 (24 July) was another suborbital splashdown. Flight 14 — slipped toward late September, FCC items still live as of 18 September — was the intended first orbital mission: on the order of 275 km, six orbits over about ten hours, and the first live payload, roughly 20 Starlink V3 satellites. Neither stage was scheduled to attempt a catch on that flight. Pad 1 at Starbase is in a deep refit into 2027, so near-term cadence is serialized through one pad. Heat-shield tiles and full-energy re-entry still decide whether Starship is a commercial truck or still a prototype. V3 deployment is a dress rehearsal for something Starwind-shaped. It is not a Starwind deployment.



A catch attempt at Starbase. Heat shield, cadence, and a working orbit are still the open items — not a million-sat compute fleet.

An A11 craft does not fly on Falcon 9. Starlink V3 already does not: each V3 is about two tonnes and the wrong shape for a Falcon fairing. Flight requirements therefore include a Starship that reaches orbit, deploys large unfolded structures, and flies often enough that a five-year satellite life does not consume the world launch market just to replenish the fleet.

Case	Launches / year	Implied daily rate
Optimistic 5-year replenishment of a million-sat class system	~3,500 Starship flights	~10 / day
Pessimistic pack-out, or 100 GW-class company remarks	~10,000+ flights	~27 / day
2025 world record, every rocket on Earth	329 orbital attempts	~0.9 / day

Even the friendly case is a twenty-fold jump in SpaceX launch rate. Last year’s world record was 329 attempts, SpaceX flying about half. Cadence math is the first filter. It is not the only one.

Layer	Proven as of Sept 2026	Filed / promised
Launcher	Falcon 9 operational; Starship still a test series	High-cadence, multi-pad orbital truck
Sats in orbit	~9,600 Starlink broadband; laser mesh exists	Up to 1 million Starlink craft, ~4 t, 5-year life
Compute flown	Starcloud H100 demo, kW-class, thermally limited; no SpaceX GPU flown	AI1 120–150 kW; later Terafab / D3 radiation-tolerant parts
Heat rejection	Starlink and Dragon-class thermal; International Space Station (ISS) ~70 kW on ~422 m² (~166 W/m²)	110 m² liquid radiators at ~1,400 W/m² claimed
Structures	Starlink deployables at ~0.6–2 t	20 m × 70 m AI1 arrays and radiators
First Starlink flight	None	Prototype talked for late 2027

Launch is only half the flight problem. SpaceX’s own Starlink thermal job listing states the point: no capable cooling architecture has been flown at this scale. AI1 wants roughly twice the ISS heat load through about a quarter of the radiator area. Early units are described as wrapping existing NVIDIA GB300 / Rubin-class parts in SpaceX hardware; a later radiation-tolerant D3 / AI7 part is tied to the Terafab story. Starcloud’s flown H100 remains the only public GPU-in-orbit data point, and it could not run at full power because the radiator was too small.

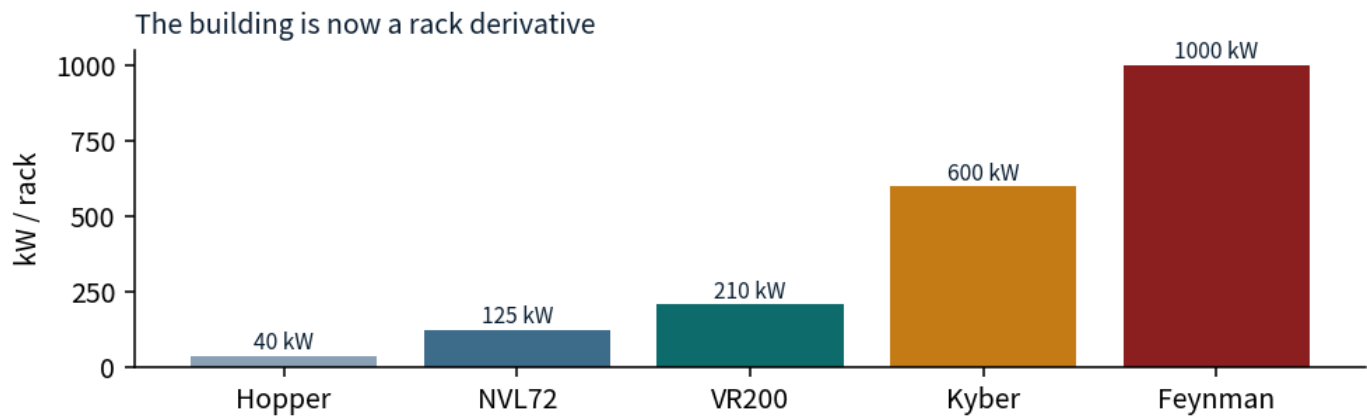
Debris, spectrum, and the 15-joule problem

Wrapping a GPU tray in radiation shielding fights the FCC’s 15-joule casualty threshold: either the craft fully vaporizes on re-entry or it needs a controlled graveyard / heliocentric disposal, which is a different congestion story. SpaceX’s September 2026 demise letter argues no surviving fragment above 15 joules. A million additional large objects also have to share shells and spectrum with Starlink and everyone else. Those fights happen in filings long before launch rate is the binding constraint. Until Flight 14 and an AI1 prototype exist as data, treat orbital GW as a filing, not a supply plan.

7. NVIDIA — the company that writes the hall

Forget the facade. Walk to the rack. NVIDIA still publishes the drawing most new halls are poured around, and that drawing keeps getting hotter — tens of kilowatts, then a hundred, then two, with six hundred on the next slide. Everyone else is trying to offer a second drawing, or to steal inference units without stealing the spec. The awkward part for lenders: the drawing changes faster than a fifteen-year loan.

Land and power decide where a hall can exist. The rack decides what the hall is. NVIDIA still writes that drawing for most of the industry. Jensen Huang has cited on the order of a trillion dollars of Blackwell and Rubin system visibility through 2027. That is a demand signal, not a delivery schedule, but it is why cooling OEMs, busbar vendors, and 800 VDC work all chase one partner blueprint.



Approximate rack envelopes in public partner specs and roadmaps. Treat Feynman-class 1 MW as a drawing, not a 2026 stock-keeping unit.

Read the next numbers as the reason the building has to be redesigned. A rack is the tall cabinet of computers. Ten years ago many halls blew cold air at cabinets drawing about 40 kW — a large home's worth of power in one closet. Hopper-era air halls lived near that 40 kW. The GB200 NVL72 — a liquid-cooled cabinet tying 72 accelerators together — pushed the industry to ~120–130 kW liquid. Vera Rubin VR200 is specified around 190–230 kW, fanless, direct-to-chip, ~45°C inlet. Kyber is drawn near 600 kW for 2027. A one-megawatt-class Feynman rack is the next slide. NVLink 6, NVIDIA's in-rack fabric, makes the rack one accelerator; Spectrum-X / InfiniBand make the hall one cluster; NVIDIA's CUDA software platform makes the software stay. 800-volt direct-current (VDC) busbars start to matter once the rack leaves the 200 kW band. NVIDIA's DSX / MaxLPS software tries to raise tokens inside a permitted megawatt — claims on the order of 40 percent more useful GPUs in the right room — because a one-gigawatt factory does not become two gigawatts without a new plant.

The real-estate consequence is ugly and under-modeled. When density jumps from ~120 kW to ~600 kW inside two years, a hall built as a fifteen-year asset starts to look like a three-to-five-year industrial envelope. That is accelerated depreciation wearing a cooling-tower costume. Older air halls and even first-generation 120 kW liquid halls are one generation from stranded unless they can be replumbed.

The financing mismatch is the part that does not show up on a cooling diagram. Standard project-finance and data-center real-estate debt is still often written on a 15- to 20-year amortization. NVIDIA's public platform cadence — Blackwell to Rubin to Kyber, with Feynman on the slide after that — is measured in something closer to 12 to 24 months per generation. A campus whose electrical room, floor load, and coolant-distribution units (CDUs) were poured for 120 kW liquid can need a retrofit long before the loan is a third of the way through. Institutional platforms talking about AI factories as an asset class (BlackRock, Blackstone, Brookfield, KKR and peers) are underwriting long paper against short physics. That is not a prediction that any named fund blows up. It is the hidden term in “tokens per megawatt as the new cap rate”: if the hall cannot be replumbed, the concrete is a husk and the credit is long.

What NVIDIA actually sells now

NVIDIA is no longer only a chip vendor. It sells a rack that behaves like one accelerator (Vera Rubin NVL72: 72 Rubin graphics processing units and 36 Vera central processing units in public partner specs), the fabric that ties racks into a hall (NVLink inside, Spectrum-X / InfiniBand outside), the software that keeps models on that fabric (CUDA, Dynamo, DSX), and — as of August 2026 — a way for someone else to finance the factory. Partnerships announced with Apollo, BlackRock, Blackstone, Brookfield, Goldman Sachs and KKR were framed as platforms meant to mobilize on the order of \$500 billion of third-party capital. SemiAnalysis and others have also described an AI Cloud Partner backstop: NVIDIA taking multi-year GPU-rental obligations so a cluster can be borrowed against. Treat the dollar headlines as company and analyst marks, not as cash already deployed. The intent is the point. NVIDIA wants the factory to be an asset class, not a purchase order.

The manufacturing story is why the spec travels. Vera Rubin was described in mid-2026 as entering full production across a partner web — Dell, HPE, Lenovo, Supermicro, Foxconn, Quanta and peers — with partner systems talked for the second half of 2026. September 2026 MLPerf Inference v6.1 preview numbers put Vera Rubin NVL72 up to about 3.7 times GB300 NVL72 throughput on a demanding vision-language test. That is a marketing benchmark. It is also why procurement departments treat Rubin as the 2026 unit. Sovereign deals sit on the same stack: a Japan factory announced with Noetra and METI around 27,500 Rubin GPUs and 140 megawatts is NVIDIA selling a national plant, not a tray of boards.

What NVIDIA does not do is own the feeder. It does not rezoning-fight Prince William. It does not run ERCOT. It writes the drawing that those fights are about. Landscape if this holds: every serious new hall is a Rubin or Kyber derivative, tokens per megawatt is how credit committees talk, and a 2025 120 kW liquid hall is one generation from stranded unless it can be replumbed. Landscape if it slips: custom silicon and AMD take more inference units, UALink becomes a real bid spec, and NVIDIA remains the training default rather than the whole factory.

8. Amazon — the company that fills the map

NVIDIA decides what the cabinet is. Amazon decides how many cities have one. Amazon Web Services is still the largest cloud by a wide margin. In 2026 it is also the largest single capital-expenditure book in the sector. Guidance moved from about \$200 billion toward \$220 billion for the year on mid-year calls, almost all of it AWS halls, power, and silicon. Treat the exact print as guidance. The direction is not subtle. AWS also talks a backlog of customer commitments on the order of half a trillion dollars. That backlog is why Amazon will keep pouring even when free cash flow goes negative for a stretch.

Project Rainier is the named AI factory. Public descriptions put the first Indiana campus (New Carlisle) in the multi-gigawatt class, Trainium-heavy, with Anthropic as the anchor — a commitment talked as high as five gigawatts of Amazon AI chips over time and more than \$100 billion of AWS spend. A second northern-Indiana book has been described in the same family. Rainier is the tell that Amazon will own the hall when the work is frontier training, and still lease colo when the work is regional inference. Time-to-power remains a product Amazon buys from landlords as well as a product it builds.

Trainium and the dual stack

Amazon is not trying to kill NVIDIA this decade. Andy Jassy has said so in public. Reported 2026–28 GPU purchase pipelines still run to millions of NVIDIA parts. What Amazon is trying to do is stop paying list price on every token AWS serves. Trainium2 is in volume. Trainium3 has been talked as ramping through 2026 on TSMC N3, with UltraServer claims that chase Blackwell-class rack work at a lower owned cost. Inferentia covers a slice of serving. Graviton covers the boring computers. Custom silicon inside Amazon has been described on an annual run-rate in the tens of billions. Talk of selling Trainium outside AWS existed in 2026; it was early, unnamed, and capacity-constrained by Anthropic and other signed demand. Read it as a lever, not as a 2027 storefront.

Power follows the same dual. Amazon is on the Susquehanna nuclear arrangement and on X-energy paper for later. It still adds regions the old way — a metro, a colo block, a transmission ask. Landscape if this holds: more secondary metros become real AWS regions because Amazon showed up with a multi-hundred-megawatt ask; Trainium sets a second floor under inference price; NVIDIA remains the guest GPU in the same buildings. Landscape if it slips: Rainier stays an Anthropic special, Trainium does not travel, and Amazon is the company that filled the map with other people's racks.

9. Microsoft — the company that writes the power contract

If NVIDIA writes the rack and Amazon fills the map, Microsoft writes the power contract — and houses a large share of OpenAI's training demand. Calendar-2026 capital expenditure has been talked in the \$175–195 billion band depending on how leases are classified; fiscal-2026 property spend and cloud backlog (\$678 billion remaining performance obligations on one print) are the demand signal. Microsoft is the hyperscaler whose cloud revenue still more nearly covers the build. That does not make the physics easier. It makes the balance sheet less desperate.

Fairwater is the named hall family. Mount Pleasant, Wisconsin, came online in 2026 as a liquid-cooled OpenAI-weighted campus and immediately collected a neighbor problem: residents heard the plant. Microsoft cut fan speeds and added sound treatment. That is Section 12 arriving inside a flagship. Atlanta Fairwater is the second named site in the same family. Analyst projections of multi-gigawatt end-states for Wisconsin should be read as envelopes, not live feeders.

Gas this decade, nuclear on paper

The Chevron West Texas behind-the-meter gas plant — public descriptions in the high-two-gigawatt class — is how Microsoft covers the years before nuclear electrons exist. Crane / Three Mile Island Unit 1, rebranded Crane Clean Energy Center, is the first named nuclear contract in the hyperscaler book: 835 megawatts, 20 years, restart widely talked for 2028 rather than the earlier 2027 slide. One reactor does not feed a 38-gigawatt ambition some secondary reports have attached to Microsoft. It sets a template others copied. Maia, Microsoft's custom accelerator, is the owned-silicon hedge on Azure traffic. It does not replace the NVIDIA halls OpenAI trains on.

Landscape if this holds: off-grid gas then a nuclear PPA becomes normal for a frontier campus; Azure is where a lot of NVIDIA silicon lives before the building is finished; community noise and rate fights travel with every Fairwater-class site. Landscape if it slips: Crane slips past 2028, West Texas gas is the whole decade, and Microsoft is a very large NVIDIA customer with a good treasurer.

10. The rest of the field — and what the plans do if they hold

NVIDIA, Amazon, and Microsoft are the three that most change other people’s buildings. The rest still matter. This section is their levers, then the hold-or-slip table for the decade.

AMD, UALink, and the ASIC camp: dual-source the stack

The anticipated plan is not to kill NVIDIA in 2027. It is to make “NVIDIA or equivalent” a sentence a chief financial officer can say. Oracle’s 50,000-GPU MI450 Helios cluster is the proof point. Google / AWS / Microsoft / Meta ASICs are the cost-per-token valve on owned traffic. UALink is the attempt to stop scale-up from remaining a single-vendor pipe. If it works, inference diversifies and campuses are built with two cooling and fabric assumptions. If it slips, dual-source stays a purchase-order hedge.

Sovereigns, SpaceX, colo

Abu Dhabi’s Stargate-class campus, other Gulf books, and French and Indian national hubs are not trying to out-invent CUDA. They are trying to make sure the next decade’s training runs can happen on their soil. That only works if power, cooling, and talent arrive on the same calendar as the chips. SpaceX is running three nested plans — Colossus speed, Starmind later, Terafab if the foundry queue is the real enemy. Colo landlords (Equinix, Digital Realty, Vantage) are trying to sell time-to-power before hyperscalers walk around them with a plant. Pennsylvania’s bring-your-own-power order, Alberta’s 1.2 GW cap, ERCOT’s audit, and New York’s environmental pause are the same sentence in different dialects.

If anticipated plans hold	If they slip
Largest clusters sit next to private generation; grid is for people	Queues and transformer waits cap new GW; announced campuses stay slides
Halls designed from Kyber / Feynman backward	120 kW liquid halls take write-downs
Inference mix shifts toward ASICs and AMD	NVIDIA keeps serving-price power
A second map for large training (Gulf, sovereign)	Those sites stay inference and fine-tune hubs
Starmind and Terafab are late, small slices of 2030 supply	Orbital GW stays a filing through the decade
Politics stays loud; barges, mines, off-grid gas remain relief valves	Same, only more so — the relief valves become the plan

Google shapes vertical silicon. TPU v7 means frontier training can live off CUDA, which is why every other ASIC program gets funded; Visakhapatnam extends the map; Project Suncatcher keeps a hand on orbital tests without owning Starmind. **Meta** shapes campus gigawatts and the nuclear book — Hyperion, Prometheus, El Paso, Sturgeon are factories next to generation, not regions, and MTIA is how Meta tries not to pay rack-tax on Llama serving. **Oracle** shapes speed and dual-source: Stargate put GB200 halls on Texas dirt faster than the industry expected from a database company, and Helios is the first public hyperscale home for a non-NVIDIA rack-scale stack.

Player	Primary lever	Landscape if they succeed
NVIDIA	Rack, fabric, software, factory finance	Every new hall is a Rubin / Kyber derivative; tokens/MW is how credit talks
Amazon / AWS	Capex + global regions + Trainium	More metros become regions; inference price has a second floor
Microsoft	Power contracts + OpenAI demand	Off-grid gas then nuclear is normal
Google	TPU vertical stack	A non-CUDA training path stays real
Meta	Multi-GW campuses + MTIA + nuclear book	Factories sit in parishes and prairie, not only Virginia
Oracle	Stargate speed + AMD Helios	A second rack-scale vendor has a reference customer
AMD / UALink	Helios rack + open scale-up	“NVIDIA or equivalent” becomes a bid specification
Colo landlords	Time-to-power shells	They remain swing capacity only where a plant has not been built

11. Capability, heat, and the HPC inheritance

A megawatt used to be a cost. Now it is supposed to be a yield — tokens, trained runs, finished simulations — and the leftover heat is supposed to be a product, not a problem. National labs learned that first, on warm water, feeding city pipes. Commercial AI is copying the lesson late, at much larger scale, and in summers where a dry cooler loses the argument.

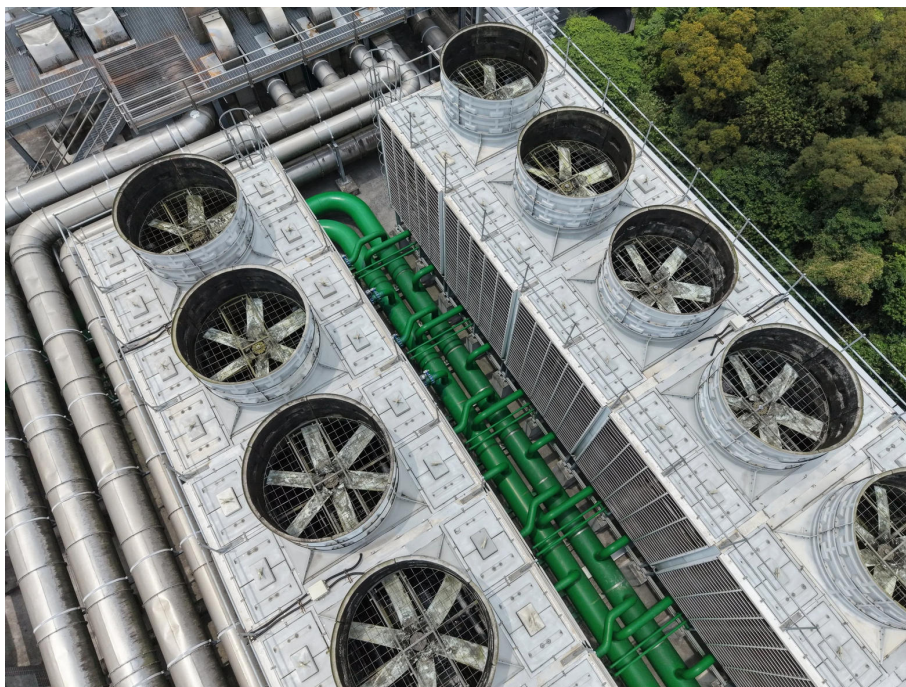
A data center used to be judged by square feet and PUE. An AI factory is judged by what a megawatt can do — tokens served, models trained, simulations finished — and by whether the heat that work produces is a waste stream or a product. Supercomputing lived with that pair of problems first. The commercial AI build is importing the answers, late and at much larger scale.

What computing capability means now

Peak floating-point operations per second (FLOPS) still appear on slides. They are no longer how operators budget a hall. NVIDIA rates a 100 MW Vera Rubin-class factory in zettaflop terms for low-precision work; the number that is starting to travel into utility and finance meetings is tokens per megawatt under a service-level objective. That metric already folds in cooling overhead, idle decode time, network stalls, and the fact that a provisioned watt is not a useful watt. HPC and AI are also converging on the same floor: Vera Rubin NVL4 blueprints at the ISC 2026 supercomputing conference are explicitly liquid-cooled units that do both FP64 simulation and AI. National machines such as LUMI-AI in Kajaani (AMD MI430X, second half 2027, an order-of-magnitude AI step over current LUMI) are being procured as factories that still have to run climate and materials codes. Capability is becoming time-varying too. Silicon Valley Power has already sent hundreds of demand signals to an AI factory that dropped load while keeping priority jobs alive. Capability is no longer a nameplate. It is what you can deliver in this hour, on this feeder, without tripping a neighbor.

Supercomputing already solved the heat problem — at lab scale

Air cooling did not fail. The rack outgrew it. Enterprise cabinets still live in the high single-digit kilowatts. A Blackwell NVL72 is ~120 kW of heat in a box; Rubin is ~190–230 kW; Kyber is being drawn at ~600 kW. You cannot blow that out with CRAC units at a temperature or noise a human hall will accept. National labs learned this on Titan, Summit, Frontier and their European twins: capture heat at the cold plate, move it in water, reject it at the plant or sell it. The design that migrated is warm-water direct liquid cooling. NVIDIA's Rubin generation is specified at up to 45°C inlet — hotter than a hot tub — and fanless end-to-end. Warm coolant is the trick, not a compromise. It lets dry coolers reject heat for most of the year, cuts or kills chillers, and drops facility water use from on the order of 2.6 million gallons per megawatt-year on cooling-tower plants toward zero in the right climate. Direct-to-chip plus a facility loop is the 2026 production path. Rear-door exchangers catch residual air load. Full immersion is real and still a minority architecture for hyperscale AI.



Evaporative plant at campus scale. Warm-water loops were meant to shrink this. In a desert they often come back as trim.

Warm water is not free in a desert

Dry coolers lose approach temperature when the outside air is already hot. In Texas, the Gulf, and the Middle East an operator either derates compute on the worst afternoons or turns evaporative trim back on — which is the water story the 45°C spec was supposed to end. Nordic heat-reuse economics do not transfer to Abu Dhabi. A chiller-less claim is a climate claim. Treat it that way in siting memos.



Desert or dry-country campus next to generation. Dry coolers work until the afternoon air is as hot as the loop.

Heat as a product

Once the loop runs at 45–65°C exit, the heat is finally hot enough to use. LUMI already feeds Kajaani’s district heating; LUMI-AI is being bought to do the same on renewable power with Bull warm-water DLC. TU Dresden’s HPC heat pumps lift server water from ~55°C to ~90°C for the city network — about 3.9 MW thermal, on the order of 3,700 households, displacing gas. OnZero’s deal with Helen in Helsinki is the AI-scale version: an “AI Heat Factory” sized around 60 MW thermal, enough in the partners’ telling for roughly 70,000 apartments, with near-zero on-site water because the heat goes into pipes instead of evaporative towers. EU Energy Efficiency Directive Article 26(6) now makes a reuse assessment mandatory above 1 MW. In northern Europe, a campus next to a fourth-generation district-heat network is a different economic object than a campus in a desert that can only reject to dry coolers or towers.



Nordic heat-reuse plant: hall plus thermal store. This economics does not transfer to Abu Dhabi or West Texas without derating or evaporative trim.

Project	Thermal note	Reuse claim (as stated)
LUMI / LUMI-AI, Kajaani	Warm-water direct liquid cooling (DLC), renewable power	Municipal district heat
TU Dresden HPC	~3.9 MW thermal via heat pumps to ~90°C	~3,700 homes; gas displaced
OnZero / Helen, Helsinki	~60 MW thermal planned	~70,000 apartments (operator claim)

12. Pushback — moratoriums, grids, communities

The buildout has neighbors now. Not a morality play about people who dislike technology — a set of fights over who pays the power bill, who drinks the cooling water, and which county or parliament will sign a twenty-year industrial neighbor. Those fights do not stop the industry. They move it.

The buildout now has a politics, and it is not a cartoon of “not in my backyard” versus progress. Project-delay trackers put on the order of 75 U.S. projects blocked or delayed in a single early-2026 quarter. Local moratoria number in the hundreds depending on how a pause is counted. Polling through 2025–26 moved from abstract support for “AI infrastructure” to local resistance once a site has a street address. That shift is the merit of the section: the constraint is no longer only turbines and GPUs. It is who pays for the feeder, who drinks the cooling water, and which county board is willing to sign a 20-year industrial neighbor.

Four drivers that keep showing up

Household rates. In PJM and Northern Virginia the argument that lands is not aesthetics. It is the claim that transmission upgrades and capacity charges for multi-hundred-megawatt campuses leak onto residential bills. Once a rate rider has a number, a data-center rezoning is a utility hearing wearing a land-use costume. That is why Loudoun and Prince William matter beyond Virginia: they taught every other market that a feeder has voters.

Water. Evaporative towers still sit on a large share of the live fleet. Operator claims on the order of 2.6 million gallons per megawatt-year are the figures communities quote back. Warm-water and dry-cooler designs were supposed to retire that fight. In drought and farm counties they have not, because peak-summer trim often brings the tower back. Water opposition is therefore not ignorance of the 45°C spec. It is disbelief that the spec will hold on the hottest week.

Land use, noise, and hours. A liquid-cooled hall is quieter than a 2018 air hall. It is still a 24-hour industrial object with truck gates, diesel or gas backup, and a night skyline. Subdivisions that were sold as residential now look at that skyline. Setback fights and lighting/noise ordinances are how those neighbors write themselves into the interconnection queue.

Tax abatements. Many sites were sold locally as jobs and tax base. Headcount per megawatt is low. When the abatement is public and the employment number is small, the political bargain collapses even if the engineering is clean. That is a different objection from rates or water, and it travels in different counties.



A lit campus next to ordinary streets. Rate riders, water, noise, and abatements travel with the feeder — not only with the rezoning map.

The notable cases — and why they are notable

Prince William / Digital Gateway is the land-use exhibit. Courts undercut rezoning for a 2,100-acre corridor. Large withdrawals followed. The lesson operators took was not “Virginia is closed.” It was “do not assemble a corridor that needs one heroic rezoning.” Subsequent U.S. campuses have preferred existing industrial land, gas-plant adjacency, or states that will pre-clear the use.

Northern Virginia / PJM is the rate exhibit. Inventory is still the world’s largest interconnection market. It is also the market where household bills became a campaign issue. New load still arrives. It arrives slower, dearer, and with political conditions that did not exist in 2019.

Texas / ERCOT is the queue exhibit. The state won the 2024–26 campus sprint because land and interconnects were faster than PJM. That success produced a ~50 GW-class queue large enough that Governor Abbott ordered a Public Utility Commission of Texas (PUCT) / ERCOT audit on 3 August 2026. San Marcos and Hill County then showed the local half of the same story: a state can want the megawatts and a town can still ban the box. Texas is not the escape from politics. It is politics arriving one cycle later than Virginia.

New York’s large-load environmental-review pause and **Pennsylvania’s** Shapiro bring-your-own-power order are the state-policy exhibits. One slows permission until the environmental file is thicker. The other says the campus may come if it brings the plant. Pennsylvania is the clearer template for the rest of the decade: not prohibition, conditionality.

Scotland’s parliamentary vote (80–26, 16 September 2026) pausing hyperscale consents pending national guidance is the European exhibit that is not Dublin-or-Frankfurt-as-usual. A national legislature treated hyperscale as a class of development that needed a rule before the next consent. FLAP-D already had grid and water stress. Scotland made that stress a vote.

Canada is the allocation exhibit, already covered in Section 5 (Canada) and repeated here because it is pushback by another name. Alberta’s 1.2 GW cap is fully spoken for; BYOP and the equipment levy are the bargain that let anything proceed. Quebec’s Bill 69 and B.C. caps are pushback from the hydro provinces that ran out of spare clean power after promising it to decarbonizing industry. The Canadian fight is not “no data centers.” It is “whose electron.”

Place	Kind of pushback	Why it matters beyond the county
Prince William, VA	Court + corridor collapse	Heroic rezonings are now treated as fragile.
NoVA / PJM	Rates and riders	Household bills became a siting constraint.
Texas / ERCOT	Queue audit + local bans	Fast interconnects attract their own politics.
New York	Environmental pause	Permission can be slowed without a ban.
Pennsylvania	BYOP order	The campus may come if it brings the plant.
Scotland	Parliamentary pause	Hyperscale treated as a national class of use.
Alberta	Cap + BYOP + levy	Megawatts proceed only on provincial terms.
Quebec / B.C.	Legislative / allocation caps	Hydro is not spare power.

Pushback does not stop the industry. It prices and relocates it. Load moves toward gas plants, secondary metros, BYOP provinces, and — in the filings — toward barges, mines, and orbit. That relocation is the bridge into Section 13 (alternate siting). It is also why transformer lead times and desert derating belong in the same conversation: a county that says yes still cannot energize a hall whose iron was not ordered years ago, and a desert county that says yes still has a water fight on the first 110°F week.

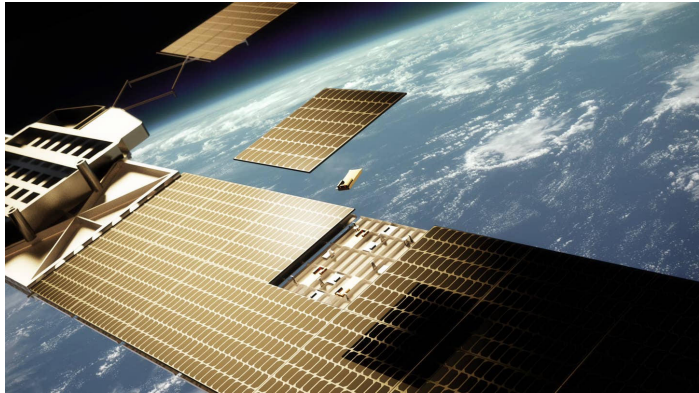
13. Alternate plans: off the parcel, off the grid, off the planet

When a county says no, or a queue says wait, the industry looks for a door that is not on the parcel. Gas on the same fence. A nuclear contract dated 2031. A barge. A sealed pod in the sea. A hall in a mine. A wind-scoured site above the Arctic Circle. A satellite. Most of those doors are real as research. Almost none of them are how a model gets trained next year.

Every alternate is a reaction to a hearing, a queue, or a transformer wait. Most of them are still small next to a single Hyperion-class campus. They matter as options and as political stories. They do not matter yet as 2027 training supply.



Underwater / sealed-pod concept. Natick is the published research; China has a commercial site. Still needs a cable.



Orbital compute as an artist's drawing. A filing and a kW-class demo — not 2027 training supply.

Approach	2026 status	Advantage	Unresolved limit
Land hyperscale	Multi-GW pouring where power exists	Scale, fiber, latency	Queues, hearings, transformers
Behind-the-meter gas	Tens of GW of equipment ordered	Skips transmission wait	Turbines sold out; emissions; gas pipes
Nuclear PPA / SMR	PPAs signed; first electrons ~2027	Firm clean baseload later	Most of the book is 2030s hardware
Floating barge / SMR	10–100 MW designs; shipyard interest	Coastal land avoidance	Maritime rules, weather, scale
Underwater pods	Microsoft Natick as research; China commercial site ~24 MW class	Ocean cooling, no towers	Service access; still needs a cable
Mines / underground	MW niches (Lefdal, Intacture and peers)	Security, rock cooling	Expansion and access
Arctic / free cooling	Narvik-class and Nordic builds	Air and hydro	Fiber, talent, winter logistics
Orbit	kW-class demo; million-sat FCC file	Solar; no zoning board	Launch rate, heat flux, demise, shells

What the cases actually prove

Behind-the-meter gas is not an alternate. It is the main escape already running. Microsoft–Chevron in West Texas, Hyperion’s plant, Alberta Greenlight, and a long turbine order book are how 2026–29 training GW gets built. The limits are the ones already in Section 3 (markets and transformers) and Section 12 (pushback): turbines sold forward into the early 2030s, gas pipes, emissions politics, and the transformer that still has to show up.

Nuclear PPAs and small modular reactors (SMRs) are the clean-firm story. The announced U.S. hyperscaler book is on the order of 10 GW. Crane / Three Mile Island (TMI) is the first named electrons , talked for ~2027. Most of the rest is 2030s hardware — Kairos, X-energy, Oklo, TerraPower, Duane Arnold restarts. Treat a signed PPA as a claim on a future plant, not as a 2026 feeder.

Microsoft Project Natick remains the best-published underwater result: a sealed pier off Orkney, power-usage effectiveness (PUE) near 1.07, no towers , and a retrieval that was a research event. It proved the thermal idea. It did not prove a service model. China’s **Hailanyun / Highlander** Shanghai site (~24 MW class in public descriptions) is the commercial data point — still a cable to shore, still a maintenance problem, still not a training factory.

Floating barges and floating SMRs are the coastal version of BYOP. Seatrrium, Samsung Heavy, and HD Hyundai have platform work; Bluecore's Long Beach floating-SMR raise is the U.S. version of the same idea. Advantage: you do not rezone a subdivision. Limits: maritime licensing, weather, and scale. A 10–100 MW barge is a regional inference node, not Hyperion.

Lefdal Mine in Norway and **Intacture** in Italy are the underground existence proof: rock cooling, physical security, and a hall that does not dominate a skyline. Expansion is the constraint. You cannot pour a 2 GW factory into a worked-out drift just because the politics are easier. **Nscale's Narvik-class** and other Arctic / Nordic builds sell free cooling and hydro. They buy fiber egress, talent, and winter logistics. They are real. They are not the default map.

Orbit is Section 6b (flight requirements) and should not be re-litigated here. Starcloud flew a thermally limited H100. Starmind is a million-sat file. Cadence, heat flux, demise, and shells are the filters. No zoning board is a real advantage. It is not a 2027 supply plan.

Every one of these is a reaction to a specific terrestrial no — a hearing, a queue, a tower, a rate rider. None of them, including Starmind, is how 2027 training runs get done. Those runs happen in a warehouse next to generation, with a transformer that was ordered years ago.

14. Bottom line

If the picture in brief was the invitation, this is the sentence you leave with. Two tracks. Four constraints. Three refinements that keep the map from lying. Then stop, unless a fact has moved.

The industry is running two tracks at once. Track one is live: halls next to existing substations, dedicated gas plants, or already-approved feeders. Track two is a press release waiting on a queue, a transformer, an environmental review, or a county vote.

Four sentences that should survive the next earnings season. Power — including the unglamorous iron of transformers and switchgear — is the constraint; land and capital are easier. The rack writes the building, and the building's useful life is shrinking as density doubles. Private generation is how serious players leave the household feeder. Orbital, underwater, and barge schemes are real research and real filings; they are not the 2027 training fleet.

Every serious player is trying to escape someone else's queue — GPUs, turbines, transformers, interconnection, or the county. Gas and NVIDIA racks are the escapes already running. Nuclear, UALink, Starmind, and Terafab are the escapes that will decide whether 2030 looks like a thicker version of 2026 or a different industry.

Three refinements sit on top of that map. Off-grid gas and remote prairie acreage solve frontier *training*, where latency is almost irrelevant; they do not retire inference in Ashburn or Frankfurt, which still has to live near users, data, and the right fiber. Interconnection queues are not a census of future halls: Capgemini's 2026 utility survey put phantom or speculative requests in front of two-thirds of electricity executives, with about 19% of those requests never becoming load, and duplicate filings trigger restudies that stall the projects that are real. And the hall-as-asset story has a tenor problem — 15- to 20-year debt against a rack cycle that can force a replumb in three to five. Layer on what the queue does not show: a site can have power and still fail an export license, a residency rule, or an insurance file. Software getting cheaper per token can make some slides optional. Announced gigawatts remain not live megawatts. Some of the announced gigawatts were never even a single building. They were the same building filed in three queues.

DATA CENTERS 2026: Illustrated Global Briefing. Joint research synthesis by Roy Poystila and Grok (xAI). This restoration dated 19 September 2026.

Glossary — shop language used in this briefing

This list is for a reader who does not live in the industry, and for a reader who does but wants the sense in which this file uses a term. It is not a textbook. Definitions follow usage in the sections above.

Units, power, and the building

Term	In this briefing
kW / MW / GW	Kilowatt, megawatt, gigawatt. A window air conditioner is ~1–2 kW. A large factory campus is hundreds of MW to several GW. 1 GW = 1,000 MW = 1,000,000 kW.
IT load	Electricity used by the computers themselves, not lights, office HVAC, or (sometimes) cooling. The number this file prefers over “number of buildings.”
First power / energize	The day a hall can actually draw its contracted load. Not groundbreaking, not a press release, not “announced GW.”
Pipeline	Announced or queued projects. A ceiling. Many pipeline gigawatts never become live megawatts.
Hyperscale	A very large campus run by or for AWS, Azure, Google, Meta, Oracle, or peers. Think 100 MW and up, standardized, often multi-building.
Colo / colocation	A landlord’s building. Tenants rent space, power, and cooling. Wholesale colo rents large blocks to hyperscalers. Time-to-power is the product.
Enterprise / edge	Enterprise: a company’s or government’s own room. Edge: a small site close to users (CDN, telco). Many sites; little of the new AI gigawatt total.
AI factory	Industry shorthand for a campus designed around high-density AI racks and liquid cooling, not a classic office-park data center.
Behind-the-meter	The campus takes power straight from a plant on the same site, instead of waiting in the public-grid interconnection queue.
Interconnection queue	The official waiting list to hook a large load or a new plant to the grid. Years long in tight markets.
GSU / transformer / switchgear	The high-voltage iron that steps power down into the building. GSU = generator step-up. Lead times of 3–4 years are why a hall next to a plant can still sit dark.
Substation	The fenced yard of transformers and switches between the transmission line and the campus.
PPA	Power purchase agreement — a long contract to buy electricity from a specific plant (often nuclear or renewables).
BYOP	Bring your own power. A government rule: the campus may come if it brings the plant. Alberta and Pennsylvania are the examples here.
SMR	Small modular reactor. Factory-built nuclear units, still mostly future hardware in this file.

Grids, markets, and agencies

Term	In this briefing
NoVA / Data Center Alley	Northern Virginia, still the world’s largest interconnection market and the cautionary tale on rates and land use.
PJM	The grid operator covering Mid-Atlantic / Midwest U.S., including Virginia. Where household-rate fights around data centers became national news.
ERCOT / PUCT	Texas grid operator / Texas utility commission. Fast interconnects pulled campuses in; the queue is now being audited.
FLAP-D	The five historic European core markets: Frankfurt, London, Amsterdam, Paris, Dublin. Now tight on power and water.
Phantom load	A grid request that is speculative, duplicated across utilities, or withdrawn. Capgemini 2026: 67% of surveyed electricity executives see it; about 19% of those flagged requests never become live load. Triggers restudies.
Restudy	Re-running an interconnection study after a project in the queue changes or drops. Often 6–18 months. The next project does not simply inherit the old result.
AESO	Alberta Electric System Operator. Ran the 1,200 MW interim cap that Meta and Keephills filled.
FCC / SAT-LOA	U.S. Federal Communications Commission; satellite license application. Starwind lives in file SAT-LOA-20260108-00016.
8-K / S-1 / 10-Q	U.S. securities filings. 8-K is news; 10-Q is quarterly; S-1 is the prospectus language this file uses for Terafab and orbital caveats.
Capex	Capital expenditure — money spent on buildings, chips, and plants, not day-to-day operating cost.

Chips, racks, and software

Term	In this briefing
Rack	The tall cabinet that holds the servers. Power per rack (40 → 120 → 200 → 600 kW in this file) is what forces the building to change.
Hopper / Blackwell / Rubin	NVIDIA product generations. Hopper is the prior air/liquid era; Blackwell (GB200 NVL72) pushed ~120 kW liquid; Vera Rubin (VR200) is the ~190–230 kW 2026 unit.
Kyber / Feynman	NVIDIA roadmap names in partner materials: ~600 kW-class, then drawings toward ~1 MW. Treat Feynman as a drawing in 2026, not a shipping SKU.
NVL72	A liquid-cooled rack that ties 72 accelerators together so software can treat them as one.
GPU / ASIC / TPU / Trainium / Maia / MTIA	GPU: general AI accelerator (NVIDIA, AMD). ASIC: custom chip. TPU is Google's; Trainium is Amazon's; Maia is Microsoft's; MTIA is Meta's.
CUDA	NVIDIA's software platform. Why most AI code still prefers NVIDIA even when other chips exist.
NVLink / UALink / InfiniBand	High-speed fabrics. NVLink ties chips inside an NVIDIA rack. UALink is the multi-vendor attempt at the same job. InfiniBand / Spectrum-X tie racks into a cluster.
Helios	AMD's rack-scale stack. Oracle's public 50,000-GPU cluster is the reference customer in this file.
CoWoS	TSMC advanced packaging used to assemble high-end AI chips. A shared bottleneck, including for NVIDIA.
800 VDC	Higher-voltage direct-current power distribution inside the hall. Less copper, less loss, once racks leave the ~200 kW band.
Training vs inference	Training builds or heavily updates a model and can sit next to cheap power. Inference answers live queries and wants to sit near users and data (data gravity).
Data gravity	Work and data attract each other. Live inference stays in fiber-dense metros; remote campuses do not cancel Ashburn.
Tokens per megawatt	Useful AI output (tokens generated or served) per unit of facility power. Starting to replace raw FLOPS in serious conversations.
DSX / MaxLPS	NVIDIA software names for squeezing more useful work out of a permitted megawatt.
HPC	High-performance computing — national-lab supercomputers. They solved warm-water cooling first; commercial AI is copying at larger scale.
FLOPS	Floating-point operations per second. Peak FLOPS still appear on slides; this file treats them as weaker than tokens per megawatt.

Cooling, heat, and water

Term	In this briefing
CRAH / CRAC / air hall	Computer-room air handlers/conditioners. The old design: blow cold air at the rack. Fine at ~40 kW. Not fine at 200 kW.
DLC / cold plate	Direct liquid cooling: water (or similar) across a plate on the chip. How modern AI racks reject heat.
Warm-water / 45°C loop	NVIDIA Rubin-generation spec: coolant arrives already hot by old standards. Lets dry coolers work and can feed city heat networks.
Dry cooler	A radiator that dumps heat to outdoor air with little or no evaporated water. Loses effectiveness when the outside air is already hot.
Evaporative tower / trim	Cooling tower that evaporates water. High water use. "Trim" means turning it back on for the hottest hours.
Thermal derating	Turning compute down because the cooling plant cannot reject the heat on a hot day. The desert problem for "chiller-less" claims.
District heat / heat reuse	Sending leftover server heat into a city pipe network. Nordic model (LUMI, Helsinki, Dresden).
PUE	Power usage effectiveness: total facility power ÷ IT power. 1.0 is perfect; older halls are higher. Less useful alone than tokens per megawatt plus water.
Chiller	Machine that makes cold water or air. Warm-water designs try to avoid it for most of the year.
CDU	Coolant distribution unit — the plant that pumps the liquid loop through the racks.

Orbit, filings, and named programs

Term	In this briefing
Colossus	xAI’s Memphis ground clusters. The only SpaceX/xAI compute layer already renting at scale.
Starmind	SpaceX FCC filing for an orbital data-center constellation of up to a million satellites.
AI1	First-generation Starmind craft in the file: large unfolded arrays, ~120–150 kW compute, ~4 tonnes.
Terafab	Proposed SpaceX/Tesla/Intel-linked silicon campus in Grimes County, Texas. S-1 still calls it a general framework.
Starship / Starlink / V3	Starship is the big reusable launcher, still in test. Starlink is the broadband constellation already flying. V3 is the heavier next Starlink generation — a dress rehearsal for Starmind-shaped cargo, not Starmind itself.
15-joule demise	FCC safety idea: no surviving fragment on re-entry should hit with more than 15 joules. Radiation shielding around a GPU fights this rule.
Shell congestion / spectrum	Too many satellites in the same orbital bands, and too many radio users on the same frequencies.
Stargate / Rainier / Hyperion / Fairwater / Crane	Named campuses or power projects: Stargate (Oracle/OpenAI Texas and G42 Abu Dhabi); Rainier (AWS Indiana); Hyperion (Meta Louisiana); Fairwater (Microsoft halls); Crane (Three Mile Island Unit 1 restart).
Natick / Hailanyun / Lefdal	Alternate-siting proofs: Microsoft’s underwater research pier; China’s commercial underwater site; a Norwegian mine hall.

Politics, in the sense this file uses the words

Term	In this briefing
Moratorium	A pause on new approvals. Local, state, or national. Not always a ban forever.
Rate rider	An extra charge on household electricity bills tied to grid upgrades. The political object in Northern Virginia / PJM.
Tax abatement	Local tax break used to attract a campus. Politically fragile when the site creates few jobs per megawatt.
§ / Sec.	Section. §5 is Section 5 (Canada). The contents table and later cross-references use the same numbers.
Peering / IX / carrier hotel	Where networks meet. Inference wants to sit near these buildings, not only near cheap power.
Export controls / residency	Rules on which chips may go where, and which data must stay on whose soil. A legal siting filter.
Insurance / climate risk	Flood, fire, hail, extreme heat. A campus that cannot buy cover on bankable terms does not close.
Announced GW are not live MW	The file’s working rule. A gigawatt on a slide is not a megawatt on a feeder.

If a term is used only once and defined in place (for example a specific bill number), it is not repeated here.

Sources

Fiber, licenses, insurance. Industry siting notes on carrier hotels and internet exchanges in Ashburn, Dallas, Dublin, Frankfurt, Tokyo. U.S. dual-use accelerator export rules and inbound investment review as legal filters on frontier clusters. Data-residency and national-cloud procurement. Insurer and lender treatment of flood, wildfire, and extreme-heat exposure on large-load campuses. Fire-code and liquid-cooling approval as a separate permit path.

Scale and pipeline. Synergy Research / TechInsights hyperscale counts (~1,297 operating, ~770 pipeline in late-cycle tallies). JLL, CBRE, Cushman 2026 outlooks. TrendForce AI power-demand series. Data Center Watch delay tallies (Q1 2026). SemiAnalysis and project trackers: treat announced GW as ceilings. Capgemini Research Institute, AI meets the grid: shaping the data center power play (June 2026): 67% of surveyed electricity executives cite phantom / speculative data-center requests; about 19% of those requests never materialize. Data Center Knowledge (June–Sept 2026) on duplicate multi-utility filings and restudy timelines. FERC large-load remarks on speculative shopping (June 2026).

U.S. siting and opposition. PJM capacity and Northern Virginia rate reporting. Prince William Digital Gateway court outcomes and subsequent withdrawals. Texas Governor Abbott PUCT/ERCOT interconnection-queue audit, 3 August 2026. San Marcos and Hill County local actions. New York large-load environmental-review pause. Pennsylvania Shapiro bring-your-own-power order. Scotland Parliament hyperscale-consent vote, 16 September 2026. Public polling on local siting, 2025–26.

Canada. The Logic national facility register. AESO large-load integration and the 1,200 MW interim cap (Meta Sturgeon 970 MW; Keephills Phase I 230 MW). Alberta government Meta Sturgeon / Greenlight Electricity Centre materials. Ontario planning outlook and Bill 40. Quebec Bill 69. B.C. large-load allocation caps.

Capex and campuses. Company 8-K / earnings guidance compiled across Q4 2025–Q2 2026. Named projects: AWS Rainier; Meta Hyperion, Prometheus, El Paso, Sturgeon; Microsoft Fairwater and Crane / TMI; Oracle/OpenAI/Crusoe Texas and G42 Abu Dhabi Stargate; Google–Adani Visakhapatnam.

Power, nuclear, gas, and iron. Public 2026 tallies of hyperscaler nuclear PPAs (~10 GW announced U.S. book). Constellation Crane / Three Mile Island restart. Amazon Susquehanna and X-energy. Google Kairos and Duane Arnold. Microsoft–Chevron West Texas behind-the-meter gas. Industry notes on GSU transformer and switchgear lead times of 3–4 years; gas-turbine order books sold forward into the early 2030s.

NVIDIA and the rack spec. GTC 2026, Hot Chips 2026, AI Infra Summit 2026: Vera Rubin VR200 NVL72, Kyber, Feynman-era drawings, NVLink 6, 45°C liquid cooling, DSX / MaxLPS, 800 VDC path. NVIDIA notes on warm-water / chiller-less operation and water-use comparisons.

Other silicon and fabrics. AMD Helios / Instinct MI450-class and Oracle's public 50,000-GPU cluster. UALink 1.0 versus NVLink. Google TPU v6/v7, AWS Trainium 2/3, Microsoft Maia, Meta MTIA 400/450/500. TrendForce CoWoS-L packaging outlook.

SpaceX, Starship, Starlink, Terafab. FCC SAT-LOA-20260108-00016 (January 2026) and September 2026 demise / 15-joule letter. SpaceX S-1 language on Terafab as a general framework and on unproven orbital technologies. AI1 design disclosures (June 2026). Grimes County filings and construction window. Starship Flight 13 (24 July 2026) and Flight 14 targeting first orbit plus Starlink V3. Cadence math from company remarks and independent pack-out ranges. Starcloud public H100-in-orbit demo. SpaceX Starlink thermal job description.

HPC, heat reuse, capability. EuroHPC / CSC LUMI-AI award (August 2026). TU Dresden HPC-to-district-heat pumps. OnZero–Helen Helsinki announcement. EU Energy Efficiency Directive Article 26(6). ISC 2026 HPC–AI liquid-cooled blueprints.

Alternate siting. Microsoft Project Natick published results. HiCloud / Highlander Hailanyun Shanghai underwater site. Seatrion, Samsung Heavy, HD Hyundai floating-platform work. Lefdal Mine (Norway) and Intacture (Italy). Nscale Narvik / Arctic free-cooling builds.

Capacities are announced or tracked. Charts are illustrative syntheses of public series. Household-equivalent and water figures are operator or project claims. Compiled 19 September 2026.